Practice workbook · Interpret human growth data, proportions and percentile charts

Height tables, body proportion diagrams and percentile growth charts from birth to adulthood.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-HUMAN-GROWTH-DATA

1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part c.ii only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

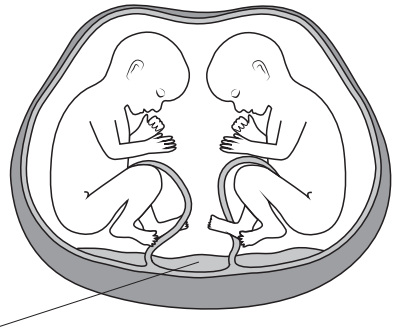
Worked-example retry · Attempt part c.ii only.

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(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

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(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part c.ii only.

(c) When babies are young, their growth is monitored using a percentile chart.

The diagram shows a percentile chart for baby girls.



(i) Determine the growth percentile of a six-week-old baby girl with a mass of 5 kg.

(1)

percentile = ...

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1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part c.ii only.

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(ii) Describe how percentile charts are used to identify concerns about the growth of a baby.

(3)

(Total for Question 8 = 13 marks)

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Working and answer

2. Question 5 · part a.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.ii only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

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mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

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2 .....

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(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

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2 .....

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2. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

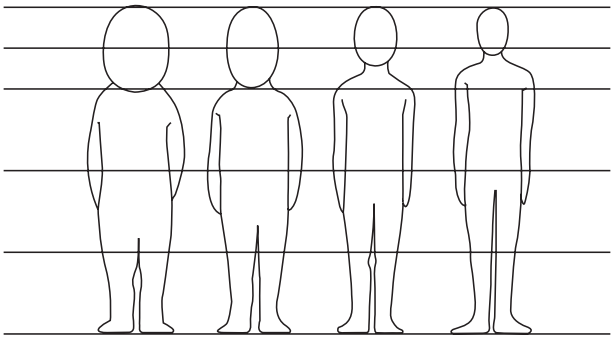
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3. Question 4 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

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2 .....

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3. Question 4 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

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(c) Suggest why it is useful to measure growth of a body part as a proportion of the whole body length.

(2)

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(Total for Question 4 = 6 marks)

Working and answer

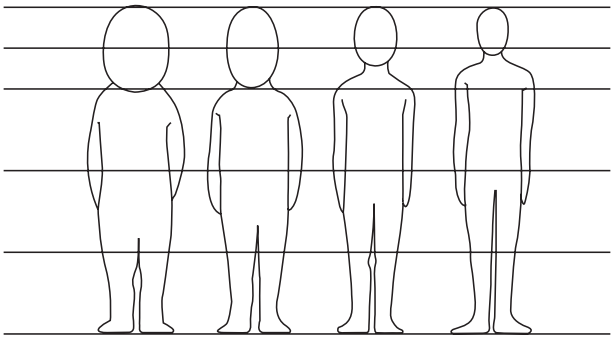
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4. Question 4 · part c · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

1 .....

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2 .....

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4. Question 4 · part c · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

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(c) Suggest why it is useful to measure growth of a body part as a proportion of the whole body length.

(2)

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(Total for Question 4 = 6 marks)

Working and answer

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